

INCIDENT REPORT

SOC Analyst Homelab Investigation

Date:	26 May 2026	Severity:	Medium
Analyst:	SOC Analyst	Status:	Resolved (Simulated)

1. Executive Summary

A simulated cyberattack was conducted in a homelab environment to practice SOC analyst skills. The attacker machine (Kali Linux, 192.168.56.101) performed multiple attack techniques against the Windows target (192.168.56.102), including PowerShell execution, credential brute forcing, and tool transfer attempts. All activities were detected using Splunk SIEM with Sysmon logging.

2. Incident Overview

Field	Details
What happened:	Attacker simulated multiple techniques against Windows target including PowerShell execution and brute force login attacks.
Account used:	Analyst / Administrator (confirmed in Splunk UserName field)
Attacker IP:	192.168.56.101 (Kali Linux)
Target system:	DESKTOP-KDJVH0Q — 192.168.56.102 (Windows VM)
Data accessed:	None — controlled simulation only

3. Timeline of Events

Time	Event	Event ID
Day 1 — Phase 1	PowerShell execution detected (T1059.001)	Sysmon EventID 1
Day 1 — Phase 2	File transfer attempt detected (T1105)	Sysmon EventID 1
Day 1 — Phase 3	Multiple failed login attempts detected (T1110)	Windows EventID 4625
Day 1 — End	All events logged by Sysmon and ingested into Splunk	—

4. Indicators of Compromise (IOCs)

Indicator	Value
Attacker IP	192.168.56.101
Target IP	192.168.56.102
Suspicious process	powershell.exe with -bypass / -encoded flags
Commands observed	IEX, Invoke-WebRequest, net use (wrong password)
Event IDs	Sysmon EventID 1 (Process Creation), Windows EventID 4625 (Failed Logon)

5. MITRE ATT&CK Techniques

Technique ID	Name	Description
T1059.001	PowerShell Execution	Malicious PowerShell commands run with -bypass / -encoded flags
T1105	Ingress Tool Transfer	File download attempted from attacker machine via Invoke-WebRequest
T1110	Brute Force	Repeated failed login attempts against the Administrator account

6. Impact Assessment

No real damage occurred as this was a controlled simulation. In a real scenario, the following risks would apply:

- Attacker could have gained initial access to the system
- Administrator credentials could have been compromised via brute force
- Malicious tools could have been installed via file transfer

7. Response Actions Taken

- Sysmon installed to capture detailed Windows event logs
- Splunk SIEM configured to collect and analyze logs from the Windows VM
- Attack activity detected and investigated through Splunk searches
- Failed login attempts identified via EventID 4625
- PowerShell abuse detected via Sysmon EventID 1 (Process Creation)

8. Recommendations

Recommendation	Action
Block PowerShell for non-admin users	Use AppLocker or Group Policy to restrict access
Enable account lockout policy	Lock account after 3 failed login attempts
Monitor EventID 4625	Set Splunk alerts for repeated failed logins in short time windows
Alert on suspicious PowerShell flags	Create detection rules for -bypass, -encoded, IEX, Invoke-WebRequest
Regular log monitoring	Use SIEM dashboards for ongoing threat visibility

This report was produced as part of a SOC Analyst homelab project. All activities were conducted in a controlled simulation environment.